

Dhvani Rakshak — Technical Hackathon Judge Q&A;

16 Short & Impressive Technical Answers (Frontend, Processing, Cryptography & AI)

■ Domain 1: Frontend Architecture & UI Tech Stack

Q1: Why did you choose React.js and Vite over Next.js or create-react-app?

Ans: Vite uses native ES Modules (ESM) and Rollup, giving us sub-second HMR and a lightweight production bundle (<300 KB) essential for low-latency dashboard streaming.

Q2: How is state managed across the 10 different feature modules without performance lag?

Ans: *App.jsx* handles active tab navigation state while each feature component manages localized hooks, preventing unnecessary re-renders across unmounted views.

Q3: How did you implement the dark glassmorphism theme without external frameworks like Tailwind?

Ans: Built a custom CSS design system using HSL color tokens (*var(--text-muted)*), CSS Grid/Flexbox, and hardware-accelerated *backdrop-filter: blur(12px)*.

Q4: How does the UI render live audio waveforms dynamically on the screen?

Ans: Using HTML5 Web Audio API *AudioContext* and SVG dynamic canvas elements to plot frequency spectrum bars in real time.

■ Domain 2: Audio Signal Processing & Feature Extraction

Q5: What exact audio preprocessing steps are applied before feature extraction?

Ans: Raw audio PCM bytes are resampled and normalized to a 16 kHz mono float32 array with amplitudes scaled between -1.0 and +1.0.

Q6: How does sliding window chunking work for live call streaming?

Ans: Audio is sliced into 500 ms sliding windows with a 100 ms hop size, producing continuous risk evaluations every 100 milliseconds.

Q7: What exact 40 acoustic features are extracted from the voice signal?

Ans: 40 Mel-Frequency Cepstral Coefficients (MFCCs), pitch jitter percentage, shimmer dB, spectral roll-off, spectral centroid, and zero-crossing rates.

Q8: How do you mathematically detect an ElevenLabs or deepfake voice clone?

Ans: AI voices exhibit artificially flat pitch jitter (< 0.05%), high-frequency phase gaps (> 4 kHz), and zero biological breathing pauses.

■ Domain 3: Security, Cryptography & Privacy

Q9: How do you enforce a 'Zero Raw Audio Retention' policy under India's DPDP Act 2023?

Ans: Audio frames are processed strictly in volatile RAM memory buffers and immediately flushed (*del raw_data*) as soon as feature extraction completes.

Q10: How are authentic voice prints stored securely during registration?

Ans: Converted into a 192-dimensional numerical vector (d-vector / ECAPA-TDNN) encrypted with AES-256, making original speech reconstruction impossible.

Q11: Why do you use SHA-256 hashing for call audit logs?

Ans: SHA-256 creates a unique 256-bit cryptographic digest for every detection event, guaranteeing tamper-evident audit logs for regulatory compliance.

Q12: How does Polygon Blockchain provide legal non-repudiation in court?

Ans: We write the SHA-256 Merkle root of the voice vector to a Polygon smart contract, creating an immutable public timestamp that cannot be edited by anyone.

■ Domain 4: Explainable AI, Performance & Scaling

Q13: How does your Explainable AI (XAI) engine calculate feature importance?

Ans: Uses **SHAP (SHapley Additive exPlanations)** to score which specific acoustic metrics (pitch, shimmer, spectrum) contributed most to raising the threat score.

Q14: How does the system handle multi-modal risk calculation?

Ans: Computes a weighted risk score: $\text{Total Risk} = 0.5 * \text{Voice Acoustics} + 0.3 * \text{Behavioral Dynamics} + 0.2 * \text{Context}$.

Q15: How would the backend scale to handle 10,000 concurrent audio WebSockets?

Ans: Using **Docker microservices**, Redis Pub/Sub async audio queues, and ONNX Runtime with TensorRT GPU model acceleration.

Q16: How do you prevent false positives when a user has a cold or sore throat?

Ans: Sore throats alter vocal timbre but **preserve natural human pitch micro-vibrations**. Borderline scores (50-70%) trigger step-up OTP checks instead of a hard call cut.